

Paper Reading Notes

Building Services and Information

Building Science / Building Information
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1 A. Building Environment and IEQ

P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks

Authors: Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, Eric Schuldenfrei

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Spatial-temporal prediction, CO2 concentration, Education facility, Space reconfigurability, BIM, Graph neural networks, Multi-modal deep learning

■**主な主張** BIM 由来の時変レイアウトと Voronoi ベースの CO2 センサ隣接グラフを multi-modal deep learning で統合し, 2 時間先予測で MAE 11.25 ppm を達成し, 比較対象中で次点の TGCN の 15.10 ppm より約 25% 低い MAE を示した. [1]

■**新規性** 静的レイアウトを前提とする既往の GNN 型 CO2 予測に対し, BIM から時変の間仕切り構成をテンソル化し, センサグラフ文脈と空間構成文脈を二枝の MMDL で統合した点に新規性がある.

■**位置付け** 教育施設の IAQ 予測を主目的としつつ, BIM 幾何情報を予測モデルの入力文脈として利用するため, Building Environment and IEQ を主分類, AI and Machine Learning for Buildings および BIM and Semantic Information を副分類として位置付ける.

P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space

Authors: Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, Jack C.P. Cheng

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

Keywords: Graph neural network, Spatial-temporal prediction, Spatial-temporal feature interactions, Indoor Air Quality, Indoor occupant, Dynamic graph learning, Occupant-in-loop

■**主な主張** 提案 AST-GCGRU は, 6 h および 12 h の multi-step prediction で graph-level R2 0.898 および 0.812 を示し, GCGRU の 0.860 および 0.699 を上回った. 12 h では occupant node R2 0.59, CO2 node R2 0.73 で, GCGRU の 0.32, 0.58 より高かった. [2]

■**新規性** 距離ベースの固定 graph ではなく, partial correlation で初期化した graph を ST-Att で学習中に更新し, time-lag nodes により 30 min CO2 sensor と 6 min occupant sensor の異なる時間解像度を保持したまま, CO2 と occupancy を同一 graph で同時予測・interaction analysis する構成を提示した.

■**位置付け** 大空間における IAQ と occupancy の同時予測・相互作用分析を主目的とするため, Building Environment and IEQ を主分類とし, AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration を副分類として位置付ける.

P005 — CO2 concentration prediction in office spaces using physics-informed neural network based on number of occupants and IoT sensor data

Authors: Jehyun Kim, Gihoon Kim, Jong-Il Bang, Anseop Choi, Minki Sung

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

Keywords: CO2 prediction, Physics-informed neural network, Occupancy, HVAC operation, IoT, Indoor air quality

■**主な主張** occupancy と HVAC/IoT data に CO2 mass-balance 制約を加えた PINN で office CO2 を予測した. [3]

■**新規性** data-driven CO2 prediction へ mass-balance equation を physics-informed loss として導入した.

■**位置付け** CO2 prediction を中心とする IEQ × PINN 研究として位置付ける.

P006 — Understanding the effect of outdoor pollution episodes and HVAC type on indoor air quality

Authors: Tristalee Mangin, Zachary Barrett, Zachary Palmer, Dillon Tang, Sean Nielson, Darrah Sleeth, Kerry Kelly

Major Category: Building Environment and IEQ

Secondary Category: HVAC and Building Services, Sensors, IoT and Data Integration

Keywords: PM2.5, HVAC, Infiltration, Wildfire smoke, Indoor air quality, Low-cost sensors

■**主な主張** 18 か月, 21 台の PM2.5 sensor で outdoor pollution episode と HVAC type による室内 PM2.5 infiltration 差を評価した. [4]

■**新規性** inversion, dust, wildfire smoke を HVAC type 別に long-term 比較した.

■**位置付け** outdoor-to-indoor PM2.5 infiltration と HVAC 影響の実測研究として位置付ける.

P014 — CO2- and PM2.5-Focused Optimal Ventilation Strategy Based on Predictive Control

Authors: Young Jae Choi, Eun Ji Choi, Jae Yoon Byun, Hyeun Jun Moon, Min Ki Sung, Jin Woo Moon

Major Category: Building Environment and IEQ

Secondary Category: HVAC and Building Services, AI and Machine Learning for Buildings

Keywords: CO2, PM2.5, Ventilation, Predictive control, VAV, Indoor air quality, Co-simulation

■**主な主張** CO2/PM2.5 予測を用いる VAV ventilation control を提案し, co-simulation で両汚染物質の制御性能を評価した. [5]

■**新規性** CO2 と PM2.5 の予測 error に応じ control priority を動的に切替えた.

■位置付け multi-pollutant IAQ predictive ventilation control 研究として位置付ける.

2 B. HVAC and Building Services

P015 — Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building

Authors: Fangzhou Guo, Sang woo Ham, Donghun Kim, Hyeun Jun Moon

Major Category: HVAC and Building Services

Secondary Category: AI and Machine Learning for Buildings, Building Environment and IEQ, Building Energy and ZEB

Keywords: Deep reinforcement learning, HVAC control, Energy, Thermal comfort, CO₂, PM_{2.5}, DDPG

■**主な主張** DDPG-based DRL で HVAC を制御し, energy, thermal comfort, CO₂, PM_{2.5} を同時最適化した. [6]

■**新規性** 複数 HVAC actuator と IEQ-energy trade-off を単一 DRL supervisory controller で扱った.

■**位置付け** HVAC supervisory control における DRL-based multi-objective optimization 研究として位置付ける.

3 C. Building Energy and ZEB

4 D. BMS, BAS, BACS and Building Automation

P011 — Integrating the Brick Schema and Control Description Language for automation and controls in BIM

Authors: Esben Visby Fjerbæk, Kevin Michael Smith, Christian Anker Hviid

Major Category: BMS, BAS, BACS and Building Automation

Secondary Category: BIM and Semantic Information, HVAC and Building Services

Keywords: BACS, Brick Schema, Control Description Language, BIM, HVAC controls, Interoperability

■**主な主張** Brick Schema と CDL を統合し, BIM 由来 HVAC equipment と control logic を machine-readable knowledge graph で接続した. [7]

■**新規性** BACS context と control sequence を Brick+CDL で一体表現した.

■**位置付け** BACS/HVAC control semantics と BIM interoperability 研究として位置付ける.

5 E. BIM and Semantic Information

P009 — A systematic comparison and evaluation of building ontologies for deploying data-driven analytics in smart buildings

Authors: Zhangcheng Qiang, Stuart Hands, Kerry Taylor, Subbu Sethuvenkatraman, Daniel Hugo, Pouya Ghiasnezhad Omran, Madhawa Perera, Armin Haller

Major Category: BIM and Semantic Information

Secondary Category: Reviews, Frameworks and Standards, Sensors, IoT and Data Integration

Keywords: Building ontology, Data interoperability, Brick Schema, RealEstateCore, Project Haystack, Digital Buildings, Ontology evaluation

■**主な主張** Brick, RealEstateCore, Project Haystack, Digital Buildings の 4 ontology を TBox/ABox 両面で比較し, alignment の必要性を示した. [8]

■**新規性** axiomatic quality と実建物 data による completeness/expressiveness を同一 framework で評価した.

■**位置付け** smart-building ontology interoperability の比較評価研究として位置付ける.

P010 — BIM-to-BRICK: Using graph modeling for IoT/BMS and spatial semantic data interoperability within digital data models of buildings

Authors: Filippo Vittori, Chuan Fu Tan, Anna Laura Pisello, Adrian Chong, Cristina Piselli, Clayton Miller

Major Category: BIM and Semantic Information

Secondary Category: BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration, Digital Twin

Keywords: BIM, Brick Schema, BMS, IoT, Knowledge graph, Data interoperability, Digital Twin

■**主な主張** BIM-to-BRICK により BIM 情報を Brick knowledge graph へ自動変換し, BIM と BMS/IoT data の semantic interoperability を実装した. [9]

■**新規性** BIM-to-Brick mapping を graph modeling で自動化した.

■**位置付け** BIM-BMS/IoT 間 semantic interoperability 研究として位置付ける.

P016 — Brick: Metadata schema for portable smart building applications

Authors: Bharathan Balaji, Arka Bhattacharya, Gabriel Fierro, Jingkun Gao, Joshua Gluck, Dezhi Hong, Aslak Johansen, Jason Koh, Joern Ploennigs, Yuvraj Agarwal, Mario Bergés, David Culler, Rajesh K. Gupta, Mikkel Baun Kjærgaard, Mani Srivastava, Kamin Whitehouse

Major Category: BIM and Semantic Information

Secondary Category: BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration

Keywords: Brick Schema, Building metadata, Semantic model, RDF, SPARQL, Portable applications

■主な主張 要検討 [10]

■新規性 要検討

■位置付け 要検討

P017 — BOT: The Building Topology Ontology of the W3C Linked Building Data Group

Authors: Mads Holten Rasmussen, Maxime Lefrançois, Georg Ferdinand Schneider, Pieter Pauwels

Major Category: BIM and Semantic Information

Secondary Category: Reviews, Frameworks and Standards

Keywords: BOT, Building topology ontology, Linked Building Data, Ontology, Semantic Web, RDF

■主な主張 要検討 [11]

■新規性 要検討

■位置付け 要検討

P018 — Semantic model-based large-scale deployment of AI-driven building management applications

Authors: Kan Xu, Zhe Chen, Fu Xiao, Jing Zhang, Hanbei Zhang, Tianyou Ma

Major Category: BIM and Semantic Information

Secondary Category: AI and Machine Learning for Buildings, BMS, BAS, BACS and Building Automation

Keywords: Semantic model, Brick Schema, AI applications, Building management, Data interoperability, Large-scale deployment

■主な主張 Brick に基づく semantic model で BIM, BMS, IoT 由来の static data と temporal data を統合し, SPARQL と FLUX による標準化された data acquisition を介して同一 application workflow を 3 建物へ展開した. 24 h ahead cooling-load prediction の CV-RMSE は Site1 14.5%, Site2 19.5%, Site3 13.5% であった. [12]

■新規性 既往の semantic model-based building applications が単一建物または open dataset 中心であったのに対し, graph database と temporal database を semantic identifier で接続する framework を構築し, educational, office, commercial の 3 実建物で deployment を実証した点に新規性がある.

■位置付け semantic information を予測モデルへの追加 feature として扱う研究ではなく, heterogeneous building data を building-independent に取得する abstraction layer として利用し, AI application の portability と large-scale deployment を支援する研究として位置付ける.

P019 — A framework for integrating BIM, building automation systems, and IoT: an application in design and operation using open standards

Authors: Shu Tang, Dennis R. Sheldon, Xinghua Gao, Cheng Zhang, Jianli Hao, Fangyu Guo, Haoyu Zhang

Major Category: BIM and Semantic Information

Secondary Category: BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration

Keywords: BIM, BAS, IoT, BOT, Brick Schema, Semantic Web, Open standards

■主な主張 要検討 [13]

■新規性 要検討

■位置付け 要検討

P021 — Metadata schema for virtual building models in digital twins: VB schema implemented in GPT-based applications

Authors: Jeyoon Lee, Sungmin Yoon

Major Category: BIM and Semantic Information

Secondary Category: Digital Twin, Retrieval, Query, LLM and Agent

Keywords: VB schema, Virtual building models, Digital Twin, Ontology, Brick Schema, GPT, HVAC

■主な主張 要検討 [14]

■新規性 要検討

■位置付け 要検討

P023 — Metadata schema for virtual sensors in digital twin-enabled building systems using Brick schema

Authors: Syed Mostasim Hasnain Saif, Jeyoon Lee, Peng Wang, Sungmin Yoon

Major Category: BIM and Semantic Information

Secondary Category: Digital Twin, Sensors, IoT and Data Integration

Keywords: Virtual sensors, Brick Schema, Metadata schema, Digital Twin, Ontology, Building systems

■主な主張 要検討 [15]

■新規性 要検討

■位置付け 要検討

P028 — Semantic Web Technologies for Indoor Environmental Quality: A Review and Ontology Design

Authors: Alex Donkers, Dujuan Yang, Bauke de Vries, Nico Baken

Major Category: BIM and Semantic Information

Secondary Category: Building Environment and IEQ, Reviews, Frameworks and Standards

Keywords: Semantic web, Linked data, Ontology, Indoor environmental quality, Building Performance Ontology, BIM, IoT

■**主な主張** IEQ modellingに必要なBIM/IoT由来の異種情報を扱うため, building topology と static/dynamic properties を統合する Building Performance Ontology を構築し, 複数 levels of detail を備えた拡張可能な semantic data model を提示した. [16]

■**新規性** 既存 ontology の systematic review から共通 design patterns を抽出し, topology, static properties, observations, actuations を lightweight upper ontology と lower-level ontologies へ統合した点に新規性がある.

■**位置付け** IEQ 向け semantic integration の基盤研究であり, BIM, IoT, BMS の異種情報を building context と結び付ける ontology 設計として位置付ける.

6 F. Digital Twin

P004 — How to measure and control indoor air quality based on intelligent digital twin platforms: A case study in China

Authors: Yuchong Qian, Jiawei Leng, Kai Zhou, Yuxuan Liu

Major Category: Digital Twin

Secondary Category: Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

Keywords: Digital Twin, Indoor air quality, BIM, IoT, Machine learning, Monitoring and control

■**主な主張** IoT, BIM, machine learning を統合した IAQ 用 digital twin platform を構築し, monitoring, prediction, visualization, control を実装した. [17]

■**新規性** IAQ の Sense-Integrate-Retrieve-Analyze-Feedback を単一 DT platform へ統合した.

■**位置付け** BIM-IoT-ML 統合による IAQ Digital Twin platform 研究として位置付ける.

P007 — A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies

Authors: Xi Hu, Rayan H. Assaad

Major Category: Digital Twin

Secondary Category: Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

Keywords: Digital Twin, BIM, IoT, Indoor environmental monitoring, Indoor positioning, Robotics, LiDAR

■**主な主張** BIM, IoT, robotics, LiDAR, UWB positioning を統合し, real-time indoor environment monitoring を DT 上で可視化した. [18]

■**新規性** mobile robotic sensing と indoor positioning を BIM-enabled DT へ統合した.

■**位置付け** BIM-enabled spatial Digital Twin による IEQ monitoring 研究として位置付ける.

P022 — Ontology-enabled AI agent-driven intelligent digital twins for building operations and maintenance

Authors: Sungmin Yoon, Jihwan Song, Jiteng Li

Major Category: Digital Twin

Secondary Category: Retrieval, Query, LLM and Agent, BIM and Semantic Information, Fault Detection, Diagnostics and Maintenance

Keywords: Digital Twin, AI agent, Ontology, Building operations, Maintenance, OpenAI API, LangChain

■主な主張 要検討 [19]

■新規性 要検討

■位置付け 要検討

P026 — Modeling indoor thermal comfort in buildings using digital twin and machine learning

Authors: Ziad ElArwady, Ahmed Kandil, Mohanad Afiffy, Mohamed Marzouk

Major Category: Digital Twin

Secondary Category: Building Environment and IEQ, AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Digital Twin, Thermal comfort, Machine learning, BIM, IoT, Indoor environment

■主な主張 要検討 [20]

■新規性 要検討

■位置付け 要検討

7 G. Sensors, IoT and Data Integration

P013 — Smart metering data enhancement in sustainable buildings via knowledge graph-guided graph neural networks

Authors: Fang He, Jiaqi Fan, Yang Deng, Xiaoyang Zhang, Ka Tai Lau, Dan Wang

Major Category: Sensors, IoT and Data Integration

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Smart metering, Building knowledge graph, Graph neural network, Data enhancement, Knowledge embedding, Brick Schema

■**主な主張** building knowledge graph と GNN を融合した BuildKnow により smart-meter data enhancement を行い, 3 実建物で評価した. [21]

■**新規性** knowledge graph 由来の meter dependence を GNN の学習 prior として利用した.

■**位置付け** smart-meter data quality と knowledge-guided GNN 研究として位置付ける.

8 H. Fault Detection, Diagnostics and Maintenance

9 I. AI and Machine Learning for Buildings

10 J. Retrieval, Query, LLM and Agent

P012 — An ontology-aided, natural language-based approach for multi-constraint BIM model querying

Authors: Mengtian Yin, Llewellyn Tang, Chris Webster, Shen Xu, Xiongyi Li, Huaquan Ying

Major Category: Retrieval, Query, LLM and Agent

Secondary Category: BIM and Semantic Information

Keywords: Natural language query, BIM retrieval, Ontology, Semantic parser, SPARQL, IFC, Multi-constraint query

■**主な主張** ontology-aided semantic parser で multi-constraint natural language query を IFC-compliant SPARQL へ変換し, 225 NLQs で 91.11% の overall accuracy を報告した. [22]

■**新規性** 複合 NL query を ontology を介して structured BIM query へ変換した.

■**位置付け** 自然言語による BIM retrieval/query 研究として位置付ける.

11 K. Occupancy and Human-Building Interaction

12 L. Reviews, Frameworks and Standards

P003 — Digital twin applications for indoor environmental quality in healthy building research: a systematic review of indicators, methods and computational paradigms

Authors: Alessandro D’Amico, Edoardo Currà, Philomena M. Bluysen

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin, Building Environment and IEQ, BIM and Semantic Information

Keywords: Digital Twin, Indoor environmental quality, Healthy buildings, BIM, IoT, Systematic review

■**主な主張** 62 件の DT × IEQ 研究を systematic review し, building semantics と occupant-related data の統合不足を整理した. [23]

■**新規性** DT × IEQ を指標, 方法, computational paradigm の観点から横断整理した.

■**位置付け** DT × IEQ の研究ギャップを整理する systematic review として位置付ける.

P008 — Digital twin-enabled smart facility management: A bibliometric review

Authors: Obaidullah Hakimi, Hexu Liu, Osama Abudayyeh

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin, BIM and Semantic Information, Fault Detection, Diagnostics and Maintenance

Keywords: Digital Twin, Facility management, BIM, Semantic interoperability, Predictive maintenance, Autonomous control

■**主な主張** 248 件を bibliometric review し, DT-enabled FM の主要潮流と semantic interoperability, predictive maintenance 等の将来課題を整理した. [24]

■**新規性** DT-enabled FM 研究の構造と future needs を bibliometric に体系化した.

■**位置付け** Digital Twin × facility management の bibliometric review として位置付ける.

P020 — Ontology-driven energy management in smart buildings: A comprehensive review of methodologies, tools, and challenges

Authors: Adamantios Bampoulas, Kaiwalya Raj, Anandhi Parthiban, Mohammad Saffari, Eleni Mangina

Major Category: Reviews, Frameworks and Standards

Secondary Category: BIM and Semantic Information, Building Energy and ZEB

Keywords: Ontology, Building energy systems, Semantic interoperability, Knowledge representation, Brick, SAREF, Digital Twin

■**主な主張** 81 件の primary study を対象に ontology-driven BES を横断整理し, ontology は standardised data flows と multi-agent coordination を支援する一方, domain coverage, emerging industry standards との alignment, predictive modelling 向け time-series integration にギャップが残ると示した. [25]

■**新規性** occupant behaviour, IoT sensor data, demand response 等に限定されていた既往 review に対し, DT integration, occupant-centric control, semantic interoperability, AI-driven optimisation を含む BES 全体の統合的 review を提示した.

■**位置付け** ontology と AI/ML, Digital Twin, real-time energy management の接続を俯瞰する review であり, semantic model を prediction/control へ接続する研究課題の上位背景として位置付ける.

P024 — Commercial building indoor environmental quality models: A critical review

Authors: Soheil Roumi, Fan Zhang, Rodney A. Stewart, Mattheos Santamouris

Major Category: Reviews, Frameworks and Standards

Secondary Category: Building Environment and IEQ

Keywords: Indoor environmental quality, IEQ models, Commercial buildings, Critical review, Thermal comfort, Indoor air quality

■**主な主張** 要検討 [26]

■**新規性** 要検討

■**位置付け** 要検討

P025 — Machine Learning and Deep Learning Methods for Enhancing Building Energy Efficiency and Indoor Environmental Quality - A Review

Authors: Paige Wenbin Tien, Shuangyu Wei, Jo Darkwa, Christopher Wood, John Kaiser Calautit

Major Category: Reviews, Frameworks and Standards

Secondary Category: AI and Machine Learning for Buildings, Building Environment and IEQ, Building Energy and ZEB

Keywords: Machine learning, Deep learning, Building energy efficiency, Indoor environmental quality, Thermal comfort, Indoor air quality

■**主な主張** 要検討 [27]

■**新規性** 要検討

■**位置付け** 要検討

P027 — Developing an integrative framework for digital twin applications in the building construction industry: A systematic literature review

Authors: Wuyan Long, Zhikang Bao, Ke Chen, S. Thomas Ng, Ibrahim Yahaya Wuni

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin

Keywords: Digital Twin, Building construction, Systematic literature review, Integrative framework, Construction, Operation and maintenance

■主な主張 要検討 [28]

■新規性 要検討

■位置付け 要検討

P029 — Operational digital twins in the built environment: A systematic review of semantic infrastructure and deployment practice

Authors: Chady Elias, Isabella Hutchins, Raja R. A. Issa

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin, BIM and Semantic Information, Retrieval, Query, LLM and Agent

Keywords: Digital Twin, Semantic interoperability, Knowledge graphs, Facility management, Systematic literature review, AI agents

■主な主張 46 件の operational Digital Twin deployment を systematic review し, formal semantic model は 25 件, semantics を runtime system logic で使用したのは 12 件であった. Runtime で semantics を用いる deployment では diagnostic output が最も多い一方, execute-level action は主として site-specific な fixed mapping に依存していた. [29]

■新規性 既往 review が definition, application, analytics 等を中心に整理してきたのに対し, semantic tier, actionability, workflow integration を用いて実装済み operational DT を分類し, semantic infrastructure の runtime 利用と deployment readiness を implementation-centered に評価した点に新規性がある.

■位置付け operational DT における semantic infrastructure を portable analytics および agent-enabled workflows の前提となる lifecycle information requirement として整理した review であり, scalable and AI-ready building operation の研究ギャップを示す位置付けとなる.

13 Uncategorized

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